

Quantum DDPM

Rishi Vora

2026-04-28

1. Overview

Classical DDPM

- Gradually adds Gaussian noise to data until the data is indistinguishable from pure noise.
- A neural network is trained to reverse this noising process by learning to denoise the data step by step.
- Once trained, the model can generate new data by starting from random noise and iteratively denoising it.

Quantum DDPM

- Gradually scrambles ensemble of quantum states until it is indistinguishable from an ensemble that is drawn from a Haar measure.
- A Parametrized Quantum Circuit (PQC) is trained to reverse this scrambling process by learning to “denoise” the quantum states step by step.
- Once trained, the model can generate new quantum states by starting from random states and iteratively denoising them.

2. Forward Process

We consider the task of the generating new elements from an unknown distribution $\mathcal{E}_0$ of quantum states, provided only a finite number of samples $\mathcal{S}_0 = \{|\psi_i^{(0)}\rangle, i = 1, \dots, m\}$ from the distribution.

For example, we choose n -qubit states clustered around $|0\rangle^{\otimes n}$

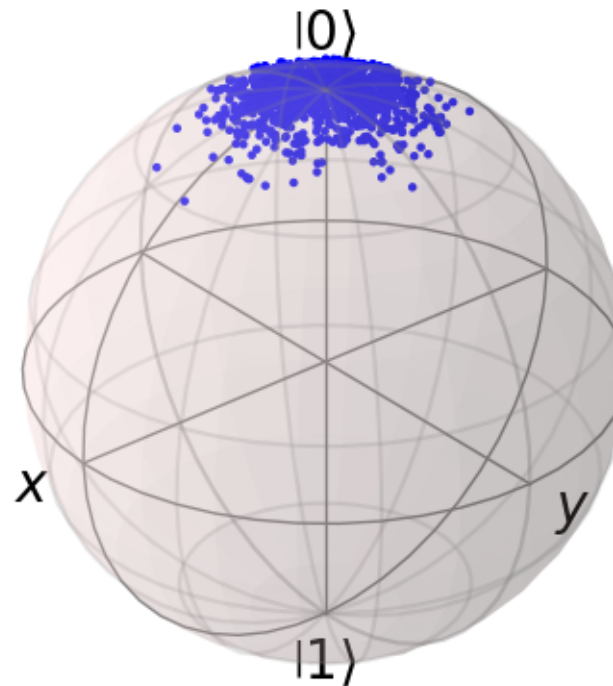
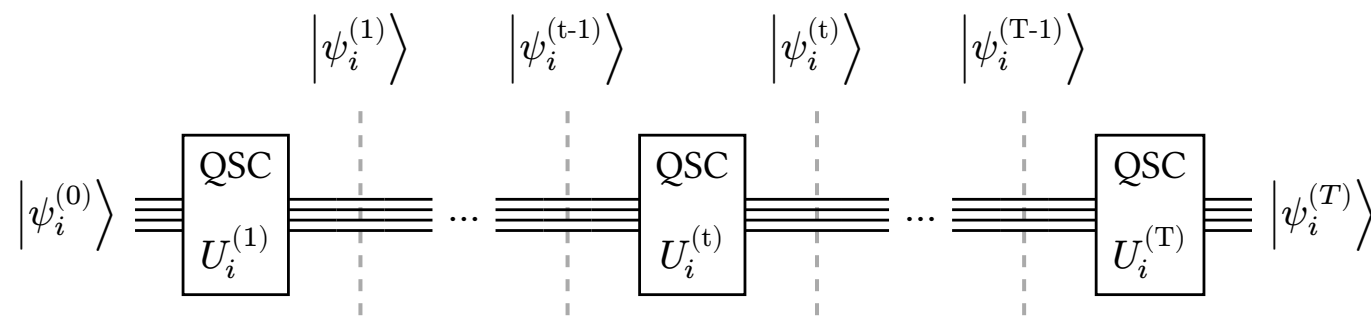
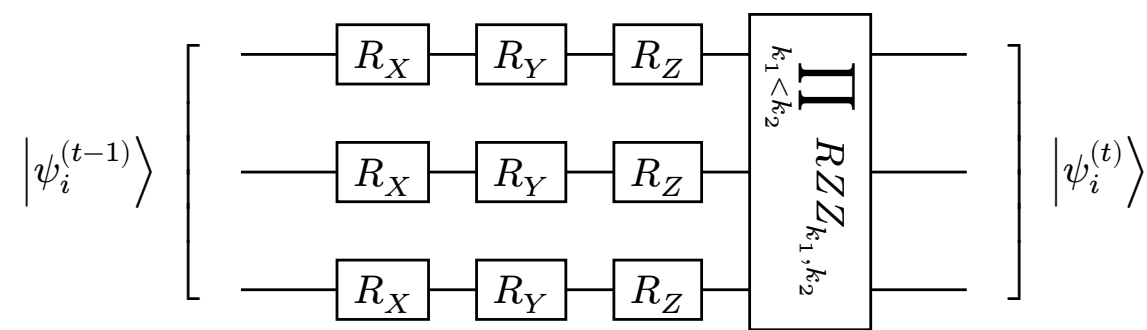


Figure 1: $m = 1000$ single qubit states clustered around $|0\rangle$

We gradually scramble it over T steps using a weight schedule β .



where each Quantum Scrambling Circuit (QSC) is like:



each R gate angle $\sim \beta_t \mathcal{U}\left(-\frac{\pi}{8}, \frac{\pi}{8}\right)$

and

each RZZ gate angle $\sim \frac{\beta_t}{2\sqrt{n}} \mathcal{U}(0.4, 0.6)$

We calculate mean fidelity of the states at each step with respect to $|0\rangle^{\otimes n}$ to quantify the scrambling:

$$\bar{F}(\mathcal{E}_1, \mathcal{E}_2) = \mathbb{E}_{|\varphi\rangle \sim \mathcal{E}_1, |\psi\rangle \sim \mathcal{E}_2} [|\langle \varphi | \psi \rangle|^2]$$

After T steps, the states $\mathcal{S}_T = \{ |\psi_i^{(T)}\rangle, i = 1, \dots, m \}$ are approximately Haar random.

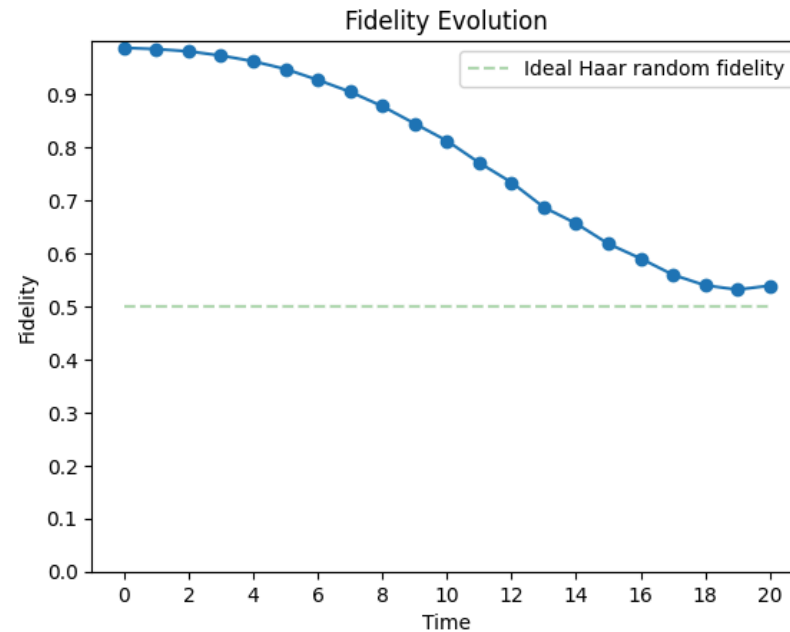


Figure 2: Gradual decrease in fidelity of single qubit states to $1/2$ over $T = 20$ steps.

3. Backward Process

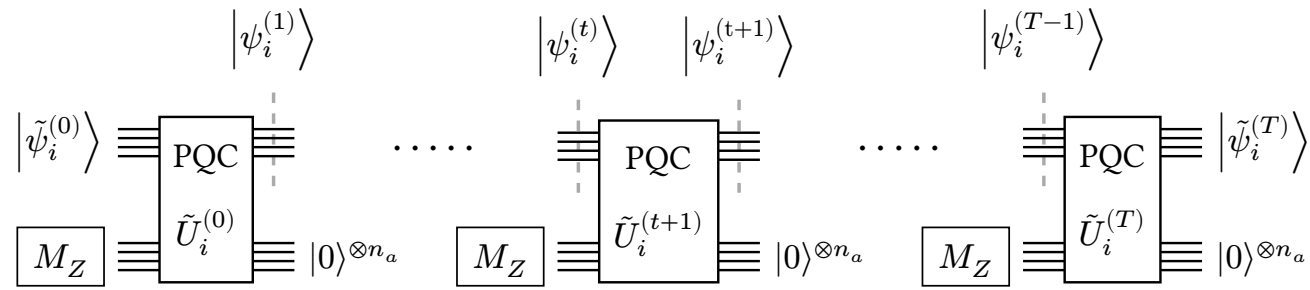
The aim is to train a series of Parametrized Quantum Circuits (PQC) to reverse the scrambling process, step by step. We start from a Haar random state $|\psi_i^{(T)}\rangle$ and apply a PQC at each step to denoise it.

Our loss function is defined as:

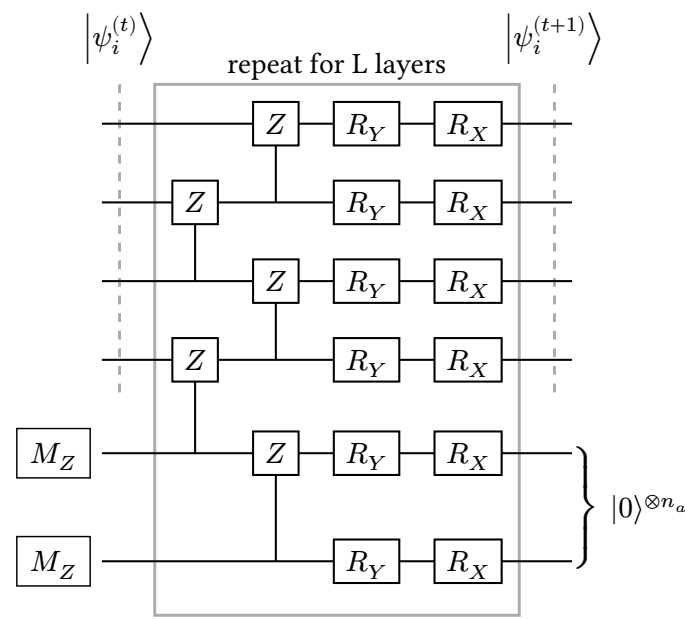
$$\mathcal{D}_{\text{MMD}}(\mathcal{E}_1, \mathcal{E}_2) = \bar{F}(\mathcal{E}_1, \mathcal{E}_1) + \bar{F}(\mathcal{E}_2, \mathcal{E}_2) - 2\bar{F}(\mathcal{E}_1, \mathcal{E}_2)$$

For practical purposes, we use our finite samples $\mathcal{S}_1$ and $\mathcal{S}_2$ to compute MMD.

We train each PQC over many epochs. In each epoch, we sample a small batch from the scrambled states $\mathcal{S}_{t+1}$ and optimize the MMD loss with respect to the target noising states $\mathcal{S}_t$.



where each PQC is like:



4. Note

Notable differences compared to the original paper:

- Their noising process was not Markovian. They were generating all states from scratch at each step.

Scopes and Limitations:

- Loss function utilizes fidelity estimation which scales poorly with number of qubits. Look for alternative loss functions.
- Learn multiple distributions and generate states according to input requesting a specific distribution.